MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL,
PRIORITY ENCODERS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/360A, dated 26 August 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* TITLE: Delete and substitute as printed above.

1.1 Scope: Delete "for each type".

1.2.3 Outline letter: Delete "X" and substitute "2".

1.3: Delete "Case E - - - - - - - - - - - - - - 50°C/W
Case F - - - - - - - - - - - - - - 70°C/W",
and substitute "Cases (all cases)- - - - - - See MIL-M-38510, appendix C". Delete "Case X" and substitute "Case 2". After 60°C/W, add "3/". After Junction temperature (T_J), add "2/".

Bottom of page add new footnotes as follows:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition per method 5004 of MIL-STD-883.

3/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein."

PAGE 2

1.4, After Case operating temperature range: Add "T_C".

3.2.3 delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request".

The attached insertable replacement pages listed below are replacements for stipulated pages. When new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	Reprinted without change
12	12

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PAGE 3

TABLE I, delete and substitute:

"

				Limits	
				Min	Max
Low-level input current at data inputs	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	-.26	-.76
0 input	I_{IL2}			-.16	-.40
Low-level input current	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	-.26	-.76
Enable input	I_{IL4}			-.12	-.36

"

TABLE I, Symbol I_{IH2}, Limits max: Delete "200" and substitute "20". I_{IH3} Limits max: Delete "20" and substitute "200".

PAGES 4 AND 5

* TABLE I, Conditions column: Delete "C_L = 50 pF ±10%" and substitute "C_L = (see figure 5)".

PAGE 5

3.6, delete and substitute: "Marking. Marking shall be in accordance with MIL-M-38510."

PAGE 6

TABLE II, Group A test requirements: Add "8" to class B devices column.

4.2 Screening, c. Delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510".

PAGE 7

FIGURE 1, Case: Delete "X" and substitute "2".

PAGE 13

TABLE III, Symbol column test 16: Delete "I_{IL1}" and substitute "I_{IL3}". Test 19: Delete "I_{IL2}" and substitute "I_{IL4}".

PAGES 14, 15, 16, 17, 18, 19, 20, 21

TABLE III, Case: Delete "X" and substitute "2". Delete "Sybmol" and substitute "Symbol".

PAGE 16

TABLE III, subgroup 10: Delete "T_C = +25°C" and substitute "T_C = 125°C".

Note 1/: Delete "Case X" and substitute "Case 2".

PAGE 21

TABLE III, Note 1/: Delete "Case X" and substitute "Case 2".

PAGE 22

4.4.3, b.(3): Delete "appendix B of MIL-M-38510 and".

6.1, Notes: Delete in its entirety.

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6.4, GND: Delete "Electrical ground (common terminal)" and substitute "Ground zero voltage potential".

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no lialbility whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Air Force - 17
Army - ER
Navy - EC

Review activities:
Air Force - 11, 19, 85, 99
Army - AR, MI
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

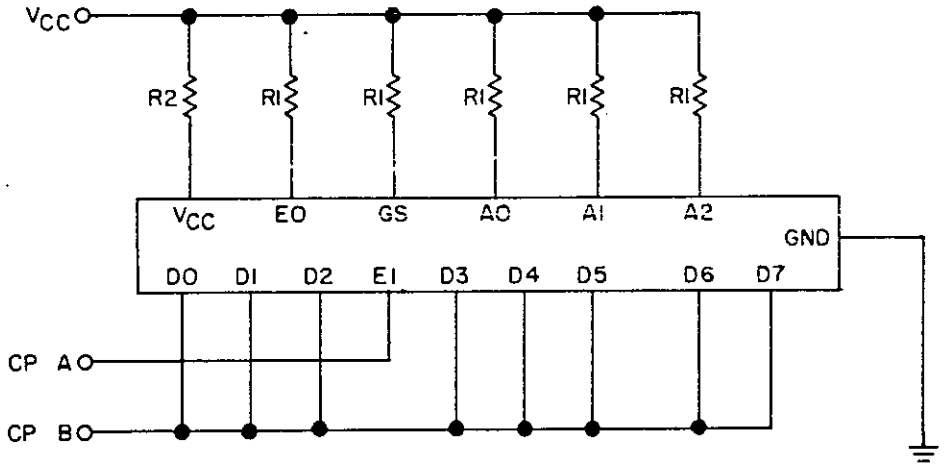
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0918)

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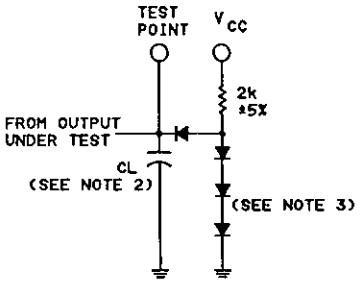
Device types 01, 02



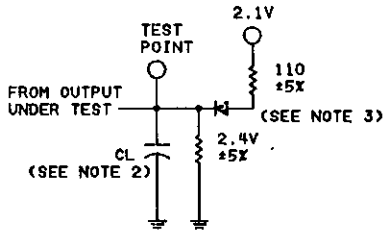
NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. RI = 1 kΩ ±5%; R2 = 10Ω max.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuit.

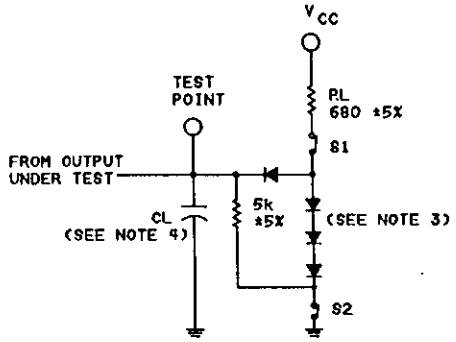
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LOAD CIRCUIT FOR DEVICE TYPE 01 (AND E0,GS,OUTPUTS FOR TYPE 02-REGULAR OUTPUT)

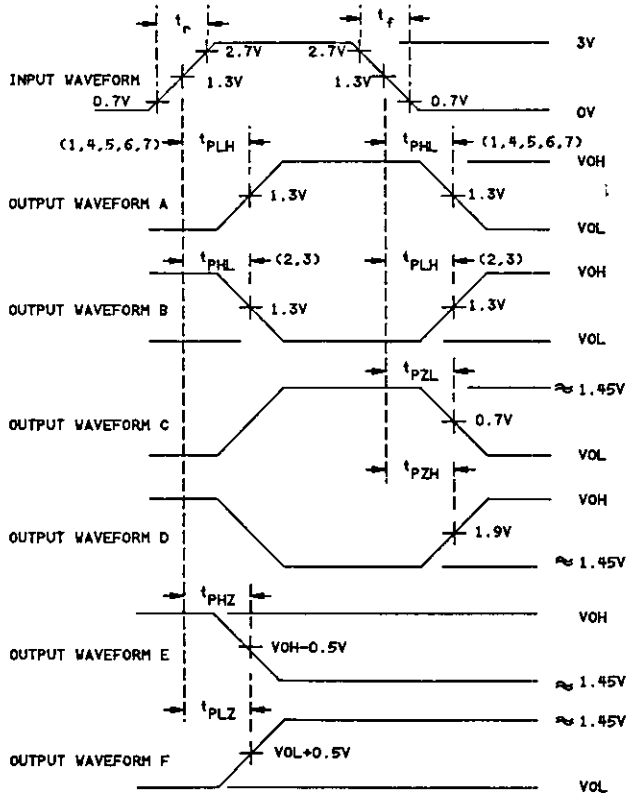


LOAD CIRCUIT FOR DEVICE TYPE 02 (A0, A1, A2 TRI-STATE OUTPUTS)

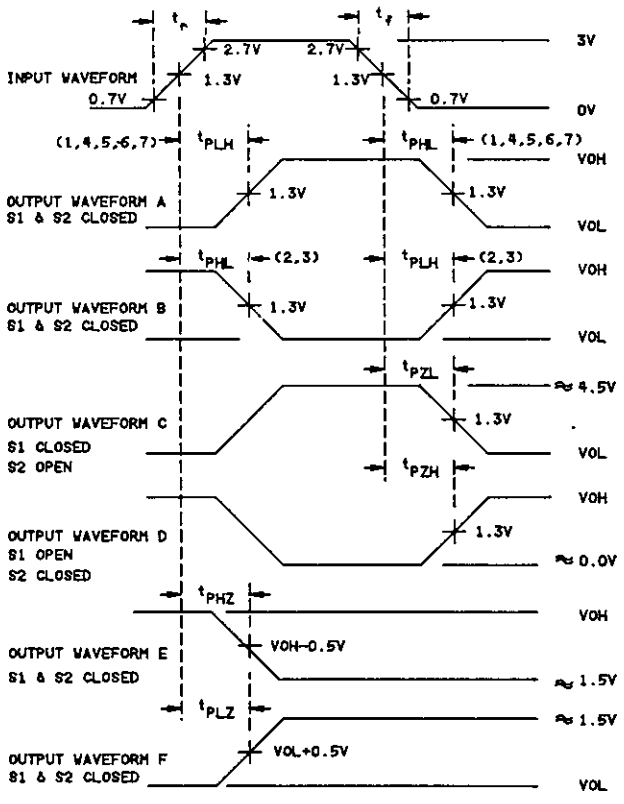


ALTERNATE LOAD CIRCUIT FOR DEVICE TYPE 02 (A0, A1,A2 TRI-STATE OUTPUT)

NORMAL AND ALTERNATE LOADS



WAVEFORM FOR NORMAL LOADS



WAVEFORM FOR ALTERNATE LOADS

NOTES:
1. Input pulse characteristics: PRR ≤ 1.0 MHz, tr ≤ 15ns, and tf ≤ 6ns.
2. For normal load Cl = 50 pf ± 10%
   Cl includes probe and jig capacitance.
3. All diodes are 1N3064 or 1N916.
4. For alternate load Cl = 50 pF ± 10% for t_{PLH}, t_{PHL}, t_{PZL}, and t_{PZH};
   CL = 15 pF min. for t_{PHZ} and t_{PLZ}.
   CL includes probe and jig capacitance.
5. Load circuits on a given output are only required where the specfic test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 5: Switching tests.

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